CSCI 111.13 Final Project Part 1: General Tutorial

Python Programming Tutorial

Learning Goals

- Students learn and understand basic Python programming
- Improve the student's programming skills through hands-on exercises

Learning Activities and Evaluation Metrics

- The student learns how each python function works
- Each lesson includes a small exercise.
    - The evaluation is based on whether the output is correct or incorrect.
    - Student just have to fill in the blanks in a code block
- There will be 5 levels, each level covers different Python lesson. The student have to pass each level

General Domain

- Programming Language (Python)
    - Proper usage and syntax of functions, variables, and statements
- Constraint-based Model
    - Will be using a constraint-based model domain
        - Step-by-Step process on what to input, with a short explanation on how this "line" of code works. Present a small example to show how it works
        - For every error made by the user, the system will provide hints.
        - Enables the student to learn through trial and error
    - **Disclaimer**: Some tasks may have different possible working lines of codes, so some tasks will have no fixed correct answer.

Specific Learning Goals

- The student will learn and understand Python programming from Print and input statements, Conditionals, Variables, and Loops.
- Student learns the fundamentals basics of python programming and knowing how the basics work.

- Student understands how to write proper code that is readable to themselves and the compiler

Criteria for Mastery

- Student finished all 5 levels
    - “A” if student finished all 5 levels with no hints used and no incorrect
    - “B+” if student only used 5 hints in total
    - “B” if student used 10 hints in total
    - “C” if the student used 15 hints in total
    - “D” if the student used all hints in total

Teaching Method

- Tutor Based
    - Minimal Feedback at first
        - Happens when committing a mistake (1-2 times)
    - Hints on what to do (the student gets this option only 3 times)
    - Review of the solution

Assessing Student Mastery

- Student finished all 5 levels correctly and with no hints
- Obtaining “A” grade

Remediating Gaps and Misconceptions

- The moment when the student still gets it wrong after exhausting all the hints. The tutorial will isolate the gap by explaining the mistake and answers the question correctly (but it will still mark as incorrect since the student was not the one who answered it)

Platforms Used

Web Browser Platform

- Will just be using Django for the website building.
    - Each level will have its own code editor with a predetermined template that is related to the level/lesson that its on.
- The AI will either be cloud-based AI using APIs or local LLM running in LM Studio or ollama
    - The AI will most likely be a free model (GPT-4o mini, Gemini 2.5 flash, Claude Haiku, Copilot)

- A local LLM is preferably used if the user's device has 8- 16 GB of RAM and 4+ GB of VRAM.